

II-Probabilistic Approaches Causality

Foundations of Neuro-Symbolic AI

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Outline

- ▶ Bayesian network hypergraphs

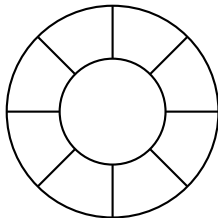
- ▶ Intervention

- ▶ Counterfactuals

Example: Random walk

Imagine a random walk through circular cells:

- ▶ X_k describes at timestep $k \in [d]$ the position
- ▶ With probability 0.5 we stay at a cell, and to each neighboring cell with 0.25



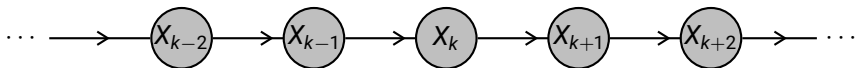
This is a Markov chain with the transition matrix:

$$\mathbb{P}[X_k | X_{k-1}] = \begin{pmatrix} 0.50 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0.25 \\ 0.25 & 0.50 & 0.25 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.25 & 0.50 & 0.25 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.25 & 0.50 & 0.25 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.25 & 0.50 & 0.25 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.25 & 0.50 & 0.25 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.25 & 0.50 & 0.25 \\ 0.25 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0.50 \end{pmatrix} [X_k, X_{k-1}]$$

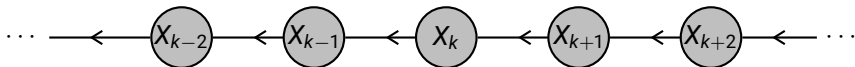
How to distinguish the past from the present?

We have two decomposition schemes as Bayesian networks, both decorated with the same transition matrices:

Parametrization from the past to the future:



Parametrization from the future to the past:



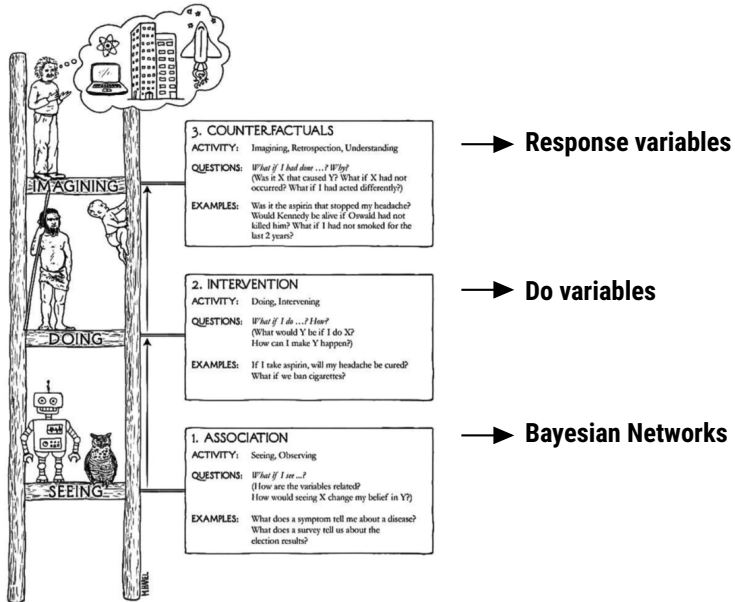
The BN hypergraph is not unique

We can manipulate the hypergraph, and still find a Bayesian network representation of a distribution.

Causation is not (always) represented in the directionality!

Idea: Distinguish by interventions.

The rungs of causation



Probability distributions: Associations

Contractions of distributions capture associations in the data:

- ▶ Marginal distributions
- ▶ Conditional distributions

Limitation: Do not model causations.

Outline

► Bayesian network hypergraphs

► Intervention

► Counterfactuals

Interventions

This appears in important situation:

- ▶ Will product sales increase if we lower the price?
- ▶ Will I pass the exam tomorrow if I study today?

Associations are not enough for these questions!

Do calculus: Modelling interventions

Add further variables D_k representing interventions:

- ▶ Intervention on the variable X_k when $d_k \in [m_k]$
- ▶ No intervention on the variable when $d_k = m_k$

This is realized by a tensor

$$\begin{aligned}\mathbb{P}[X_k | X_{\text{Pa}(k)}, D_k] &= \sum_{x_k \in [m_k]} \epsilon_{x_k} [X_k] \otimes \mathbb{I} [X_{\text{Pa}(k)}] \otimes \epsilon_{x_k} [D_k] \quad (\text{intervention}) \\ &+ \mathbb{P}[X_k | X_{\text{Pa}(k)}] \otimes \epsilon_{m_k} [D_k] \quad (\text{no intervention})\end{aligned}$$

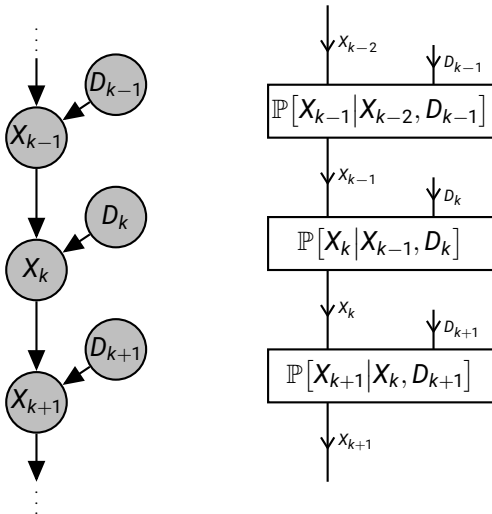
This resolves the ambiguity in Bayesian network representation!

Example: Random walk

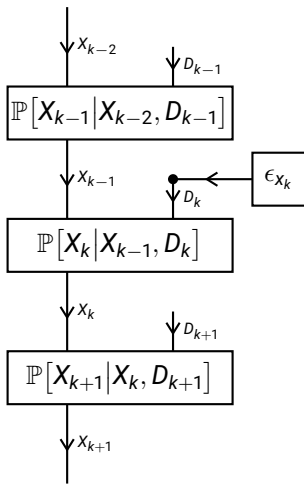
Random walk example with $m = 3$:

$$\mathbb{P}[X_k | X_{k-1}, D_k] = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0.50 & 0.25 & 0.25 \\ 0.25 & 0.50 & 0.25 \\ 0.25 & 0.25 & 0.50 \end{pmatrix} [X_k, X_{k-1}, D_k]$$

Causally augmented Markov chain

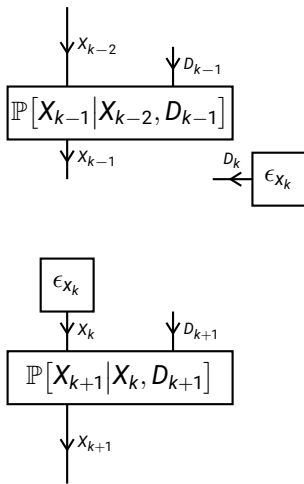


Intervening on a position



Interpretation: Restart of the random walk at time k in intervention state X_k .

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- ▶ Bayesian network hypergraphs
- ▶ Intervention
- ▶ **Counterfactuals**

Counterfactual worlds

- ▶ A particular state has been observed.
- ▶ We reason about a hypothetical world.

This appears in important situation:

- ▶ Would the accident have happened had the safety protocol been respected?
- ▶ Would the person have developed cancer had he stopped smoking earlier?

Interventions are not enough for these questions!

Response variables

Model each conditioned probability as a basis encoding of a function on the parent variables, the do-variable and the response variable

$$L_k$$

The response variables are shared with the counterfactual world variables.

Random walk example

We model a response variable

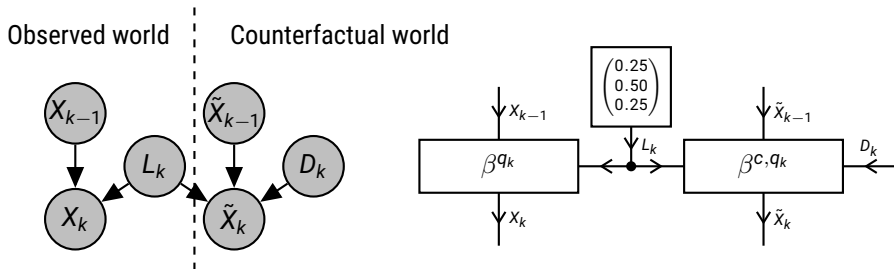
- ▶ $\ell_k = 0$: going to the left cell
- ▶ $\ell_k = 0$: staying at the current cell
- ▶ $\ell_k = 2$: going to the right cell

$$\mathbb{P}[X_k | X_{k-1}, L_k] = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} [X_k, X_{k-1}, L_k]$$

Effect in random walk: Shift of the random walk.

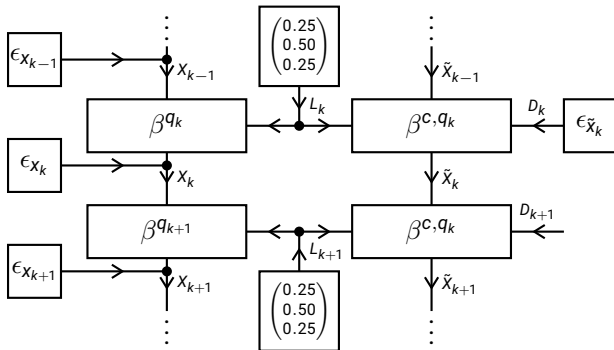
Counterfactual random walk

What will be the outcome of the walk, when we do an intervention on X_k ?



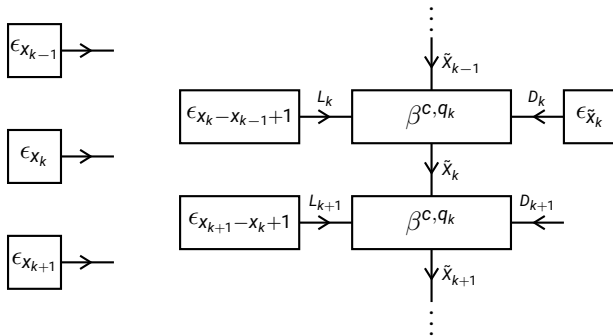
Counterfactual query

What would be the outcome of the walk, would we have done the intervention on X_k ?



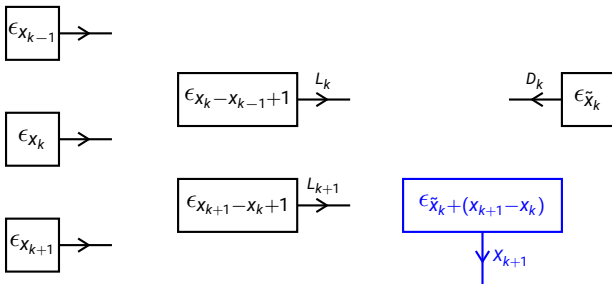
Counterfactual query

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Counterfactual query

What would be the outcome of the walk, would we have done the intervention on X_k ?



Answer: The walk would be shifted by the intervention. Different to the intervention query, we have a deterministic outcome!